

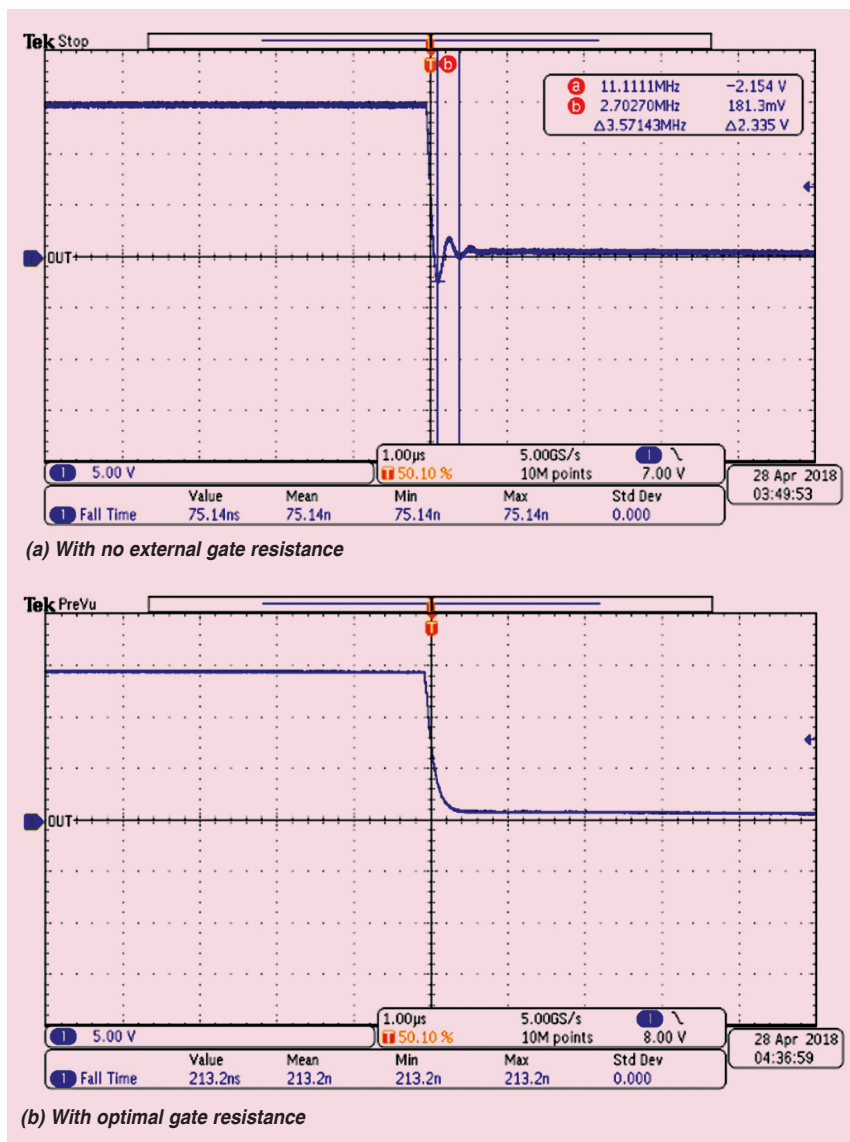


Fig. 3: Impact of external gate resistance on overshoots (Source: TI Tech Note 'External Gate Resistor Design Guide for Gate Drivers')

Wide-bandgap semiconductor devices operate at much higher frequencies than silicon based devices. The faster switching speed means higher dv/dt (change in voltage with respect to time) and di/dt (change in current with respect to time), which leads to greater oscillation challenge.

Various kinds of oscillations arise under different circumstances, but one of the prominent kinds is sustained oscillation of the drain-source voltage. This problem occurs only in GaN transistors as they do not incorporate body diodes. When the gate-source voltage or the gate-drain voltage is higher than the threshold voltage of the GaN device, the electrons are attracted to

the heterostructure interface of the aluminium gallium nitride (AlGaN), and GaN devices conduct reverse current under such conditions. The sustained oscillation may occur due to the reverse conduction characteristics of GaN devices and parasitic parameters.

The switching oscillations can cause many undesired negative effects, including voltage and current overshoots, shoot-through, EMI, additional power loss, and even device damage, which can seriously affect the performance of power converters and systems. Voltage and current overshoots are common in GaN devices during turn-on and turn-off transients. For a device operating at

600V, the voltage overshoots can peak at 800-900V, resulting in a catastrophic failure.

In addition, high dv/dt and di/dt may cause the gate-source voltage to exceed the threshold voltage, thereby inducing false turn-on. When false turn-on occurs, the power loop inductance has an input voltage across it, and therefore a large short-circuit current will pass through the two switches. From various experimental results, the short-circuit current rises to nearly 150A, which seriously affects the performance and reliability of the system.

The high dv/dt and di/dt , moreover, cause severe conducted and radiated EMI noise. Conducted and radiated EMI emissions often produce undesirable effects on electronic systems such as radio receivers, medical equipment, and communication systems. Besides, they cause malfunctions and non-operations in control systems.

Apart from degradation in performance, the oscillations in GaN circuits can also cause additional power loss. There are various conventional and unconventional ways, apart from active gate drivers, to suppress these issues occurring due to high dv/dt and di/dt .

Following are some of the design and PCB layout tips to reduce these effects and their impact on the overall device performance.

Design tips for gate driving

1. Switching speed control. Switching speed is limited by the low-pass filter formed at the gate due to the gate resistance and gate capacitances. One can easily control the switching speed by adding an external gate resistor. The external gate resistor will increase the time constant associated with the pole and decrease the maximum switching speed. On the other hand, it will play a crucial role in limiting noise and ringing in the gate drive path. It is observed that the parasitic inductances and capacitances, high dV/dt and di/dt , and body-diode reverse recovery can cause unwanted behaviour without an appropriately sized gate resistor.

Fig. 1 depicts the common gate elements in the gate drive path: the internal resistance of the gate driver,